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Immunomodulatory effects of hydroxyethylated Hericium erinaceus polysaccharide on macrophages RAW264.7

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Abstract:

Hericium erinaceus polysaccharide (HEP) has been shown to possess a variety of biological activities. In present study, HEP was successfully modified to obtain its hydroxyethylated derivative hHEP. Its potential immunomodulatory activities on RAW264.7 macrophages were investigated. Results showed that the hHEP were significantly stronger than that of the corresponding unmodified polysaccharide, HEP. Meanwhile, the NO, IL-6 and TNF-α production activities of macrophages were enhanced in the RAW264.7 macrophages by stimulation of hHEP. In addition, the hHEP increase significantly higher iNOS expression than HEP. These results indicated that the hydroxyethylated derivative hHEP could enhance the activation of peritoneal macrophages, and hydroxyethylation modification can enhance the immunomodulation function of HEP.

Keywords: Hericium erinaceus polysaccharide; Hydroxyethylation; Immunomodulation

1. Introduction

Hericium erinaceus (Bull.:Fr.) Pers. has been used as functional foods and herbal folk medicine in China for a long time [1-3]. It belongs to the family Hericiaceae. It is a traditional medicinal food for the prevention and treatment of chronic atrophic gastritis (CAG), duodenal ulcers and several other diseases [2]. Several classes of phytochemicals, such as polysaccharides, ketones, terpenoids and phenolic compounds have been isolated and identified from the H. erinaceus. [4-7]. The

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polysaccharides derived from H. erinaceus is a main active ingredient, and has various biological activities such as anti-tumor, anti-oxidation, antimicrobial, anti-inflammatory and immunomodulatory [8-12].

Polysaccharides is a kind of important biological macromolecules, but their bioactivities are not always in the condition of satisfaction [13,14]. Therefore, molecular modification of polysaccharide is an significant means to enhance its biological activities. After modification, the inactive or less active polysaccharide is greatly improved. As an effective chemical modification method of polysaccharide, hydroxyethylation is interesting due to operate simple and its significant effects on improve the physiological properties [15]. Many researchers have demonstrated that hydroxyethylated modification polysaccharides were significantly improved their immune activity [16,17]. However, the hydroxyethylated modification influence on the immunoregulatory effects of polysaccharides from H. erinaceus remains unclear.

Macrophages is one kind of important immune cells and the body's first line of defense against infection. It play a critical role in the innate and adaptive immune responses [18]. Under normal conditions, macrophages are in the static state. But when the body is stimulated by pathologic stimuli or injury, macrophages could kill these foreign matters indirectly via the secretion of macrophage-derived biological factors, including nitric oxide (NO), iNOS, reactive oxygen species (ROS), and several cytokines such as IL-6, IL-8, TNF-α and so on [19, 20]. Thus, macrophages have been broadly used to evaluate the immunomodulatory potential of bioactive polysaccharides through multiple parameters analysis.

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Based on the previous studies [21], we prepared the hydroxyethylation derivative of H. erinaceus polysaccharide and its immunomodulatory activities were evaluated and compared. The immunomodulatory effects of hydroxyethylated modification polysaccharides from H. erinaceus on RAW264.7 macrophages were determined by assessing NO, IL-6, TNF-α production, iNOS expression and phagocytosis ability. The aim is to obtain a good immune activity of the hydroxyethylation polysaccharide from H. erinaceus, and provide theoretical basis for further clinical application of immunologic adjuvant and immune enhancer.

2. Methods and materials

2.1. Materials and chemicals

H. erinaceus was obtained from Huishihao Pharmacy in Fujian Province of China. The identities of the plants were verified by Dr. Yifan Huang and deposited (voucher no. 20160715) in Key Laboratory of Traditional Chinese Veterinary Medicine and Animal Health in Fujian province, Fujian Agriculture and Forestry University.

RAW264.7 cells were obtained from institute of cell biology, Chinese academy sciences (Shanghai, China). DMEM was purchased from Gibco (Invitrogen Corporation, U.S.). 3-(4,5-Dimethylthiazol-2-yl)-2,5-diphenyltetrazolium bromide (MTT), Dimethyl sulfoxide (DMSO) and Lipopolysaccharide (LPS) were purchased from Sigma Co., USA. Assay kits for interleukin-6 (IL-6), tumor necrosis factor-alpha (TNF-α), nitric oxide (NO) test kits and Coomassie Brilliant Blue G-250 were all obtained from Nanjing Jiancheng Bioengineering Institute (Nanjing, Jiangsu Province,

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China). FITC-dextran was purchased from BD Pharmingen (San Diego, CA, USA). Rabbit anti-iNOS monoclonal antibody, HRP conjugated anti-rabbit antibody and rabbit anti-β-actin were obtained from Santa Cruz Biotechnology (California, CA, USA). All other chemicals and solvents were used of analytical reagent grade and obtained from Sinopharm Chemical Reagent Co., Ltd. (Shanghai, China).

2.2. Hydroxyethylated modification of HEP

The purified H. erinaceus polysaccharides (HEP) were prepared as previously reported [21]. Hydroxyethylated modification of HEP was prepared according to the method reported by Zhang and Tian [16]. Briefly, 500 mg of HEP was added respectively into the three-necked flask filled with 20 mL of the mixture (1 % NaOH and 3 % NaCl solution) stirring to make HEP completely dissolve. Under the condition in 0 ℃, the mixture was added and stirred 2 mL of epoxyethane at 2 h. Then the mixture was recovered to room temperature and mixed for 24 h. Afterward, the next day the mixture was adjusted pH to 7 with 1 $mol \cdot L^{-1}$ HCl. The precipitate was obtained by adding 4 to 5 times the volume of acetone followed by centrifugation, and the final hydroxyethylated derivative products, named as hHEP, was obtained by dialyzed and freeze dried. Its carbohydrate content was determined by the phenol-sulfuric acid method [22].

2.3. Identification of hHEP

2.3.1. Infrared spectroscopy analysis

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The FT-IR spectra of HEP and hHEP in a wave number range of 4000-400 cm^{-1} were recorded by KBr pellets method with a Nicolet 200 Magna-IR spectrometer (Nicolet Instrument Corp).

2.3.2. Molecular weight analysis

The average M_W was measured through GPC on an ultrahydrogel linear column at 25 ℃. The eluent (flow rate: 1.0 $mL \cdot min^{-1}$) was a 0.2 M sodium acetate solution. Pullulan standards were used as the standards for M_W measurement (M_W: 6.1-70.8 kDa).

2.3.3. Monosaccharide composition

Monosaccharide compositions were analyzed by high-performance anion exchange chromatography coupled with pulsed ampero metric detection, after acid hydrolysis [23].

2.4. Cell viability assay

The effect of HEP and hHEP on the viability of RAW264.7 cells were determined by MTT assay. Briefly, RAW264.7 cells were seeded at 1×10^6 $cells \cdot mL^{-1}$ in a 96-well plated for 24 h and then were cultured with different concentrations of HEP and hHEP for 24 h. After incubation, MTT was added to each well 30 μL to a final concentration of 0.5 $mg \cdot mL^{-1}$. The plates were continued to incubate for 4 h, and then the supernatants were removed before adding DMSO (100 $\mu L \cdot well^{-1}$). Finally, the absorbance at 570 nm was measured by microplate ELISA reader.

2.5. Determination of phagocytic uptake

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RAW264.7 cells were collected at the end of the culture with HEP, hHEP or LPS, and then incubated with 1 mg·mL^{-1} FITC-dextran at 37 ℃ for 1 h. The reaction was stopped by ice-cold PBS containing 2 % FBS. The cells were washed three times with cold PBS. Cellular uptake of FITC-dextran was analyzed by BD Accuri flow cytometer.

2.6 Measurement of NO production

RAW264.7 cells were seeded at 1×10^6 cells·mL^{-1} in 96-well plates and cultured for 24 h. Then, the cells were pretreated with HEP and hHEP at various concentration or LPS (1 μg·mL^{-1}) for 24 h at 37 ℃. Then cell culture supernatants were collected, and the NO level was determined by the Nitric Oxide Assay Kit (Jiancheng, Jiangsu, China) based on the Griess reagent method [24].

2.7. Cytokine assays

RAW264.7 cells (5×10^5 cells·mL^{-1}) were seeded in 96-well plates and cultured for 24 h with the HEP and hHEP at various concentration as above. After 24 h of treatment, the supernatants were collected by centrifugation at 1000 rpm for 8 min at 4 ℃ and stored at -20 ℃. The concentrations of IL-6 and TNF-α were determined using enzyme-linked immunosorbent assay (ELISA) kits, according to the manufacturer's instructions.

2.8 Western blot analysis

RAW264.7 cells (1×10^6 cells·mL^{-1}) were seeded in 6-well plates for 24 h and pretreated with HEP and hHEP at various concentrations or LPS (1 μg·mL^{-1}) for 12 h. RAW264.7 cells were washed twice with cold PBS and lysates were prepared from

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total cells. The denatured proteins were separated on 10 % sodium dodecyl sulfate polyacrylamide gel electrophoresis (SDS-PAGE), and transferred to PVDF membrane. The membrane was blocked with 5 % non-fat milk in Tris-buffered saline containing 0.05 % Tween 20 for 1 h, and then incubated with an appropriate dilution of primary antibodies in 5 % BSA (in Tris-buffered saline containing Tween 20) for 2 h. The blot was incubated with target monoclonal antibody iNOS in 2.5 % milk-TBST solution at 4 ℃ overnight. The membrane was then washed with Tris-buffered saline containing Tween 20 and incubated with HRP conjugated secondary antibody. After washing the membrane with TBST three times for 5 min, the cells photographed under a confocal laser scanning microscope. β-actin was used as the loading control.

2.9 Statistical analysis

All data obtained in this study were processed statistically and divergence were presented as mean ± SD. SPSS 16.0 for Windows was used and $P < 0.05$ indicated significance differences.

3. Results

3.1. The infrared spectra of HEP and hHEP

The FT-IR spectra of HEP and hHEP were illustrated in Fig. 1. In the spectra of HEP and hHEP, there were three band, one appeared in 3600-3200 cm^{-1} corresponding to the hydroxyl stretching vibration, the second appeared in the 2922 cm^{-1} or 2924 cm^{-1} corresponding to methyl C-H stretching vibration, the third appeared in 1400-1000 cm^{-1} corresponding to C-O-C stretching vibration. This indicated that the HEP and hHEP were polysaccharides.

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As compared with the spectrum of HEP, the spectrum of hHEP presented one characteristic absorption bands, one appeared at 1363 cm^{-1} describing a symmetrical $-CH_2-$ stretching vibration (Fig. 1B), which indicated that hHEP was successfully modified in hydroxyethylation.

3.2. Physicochemical features of HEP and hHEP

As shown in Table 1, the chemical analysis of HEP revealed that carbohydrate content and molecular mass were 81.3 % (w/w) and 16.15 kDa, respectively. An hydroxyethylated derivative, hHEP, was prepared in this study. hHEP's carbohydrate content and molecular mass were 75.4 % (w/w) and 15 kDa, respectively. The monosaccharide compositions in hHEP were mainly composed of glucose, fructose, galactose, mannose and arabinose with a molar ratio of 30.23: 25.30: 2.45: 1.04: 0.18.

3.3. Effects of HEP and hHEP on the viability of RAW264.7 cells

After treated with HEP and hHEP (25-400 $\mu g \cdot mL^{-1}$) for 24 h, cells were detected for the viability using MTT assay. As shown in Fig. 2, compared control group, HEP at 25-200 $\mu g \cdot mL^{-1}$ and hHEP at 50-400 $\mu g \cdot mL^{-1}$ groups were not significantly larger than that of the cell control group ($P > 0.05$). To avoid the cytotoxicity of test sample, HEP and hHEP at the indicated concentrations of 50, 100, 200 $\mu g \cdot mL^{-1}$ were selected for following assays.

3.4. Effect of HEP and hHEP on NO release from RAW264.7 cells

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As shown in Fig. 3, the NO concentration of the culture supernatant of the RAW264.7 cells were significantly increased by treated with different doses of HEP and hHEP (50, 100 and 200 μg·mL^{-1}) ($P < 0.05$). Compared to HEP groups, hHEP (100 and 200 μg·mL^{-1}) were significantly increased higher NO secretion ($P < 0.05$). At the same time, the hHEP at 200 μg·mL^{-1} was significantly increased higher than those in other groups ($P < 0.05$) except for LPS group ($P > 0.05$).

3.5. Effect of HEP and hHEP on phagocytic uptake from RAW264.7 cells

As shown by flow cytometry (Fig. 4), HEP and hHEP increased the phagocytosis of the cells in a dose-dependent manner. Compared to HEP groups, hHEP (100 and 200 μg·mL^{-1}) were significantly increased higher phagocytosis of the RAW264.7 cells ($P < 0.05$). LPS was significantly increased the phagocytosis ability of the RAW264.7 cells than other groups ($P < 0.05$) except for hHEP at 200 μg·mL^{-1} group ($P > 0.05$).

3.6. IL-6 and TNF-α determination

As shown in Fig. 5, HEP and hHEP increased the IL-6 and TNF-α production in a dose-dependent manner. Compared to HEP groups, hHEP at 200 μg·mL^{-1} group was significantly increased higher IL-6 production ($P < 0.05$) (Fig. 5a). hHEP (100 and 200 μg·mL^{-1}) groups were significantly increased TNF-α production than HEP groups ($P < 0.05$) (Fig. 5b). At the same time, the hHEP at 200 μg·mL^{-1} was significantly increased higher TNF-α production than LPS group ($P < 0.05$).

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3.7. Effect of HEP and hHEP on iNOS expression in RAW264.7 cells

As detected by western blotting (Fig. 6), the protein levels of iNOS was significantly up-regulated by LPS than other groups ($P < 0.05$). Compared to HEP groups, hHEP (100 and 200 μg·mL^{-1}) groups were significantly increased higher iNOS expression ($P < 0.05$).

4. Discussion

Macrophages are important immune cells with diverse functions. Macrophages could phagocytose and kill pathogenic microorganisms, release some inflammatory mediators and active cytokines to defense against invading pathogens [25, 26]. It could also present the pathogen to the lymphatic tissue, which activating the specific immune response and regulating the body immunity [27]. Therefore, in the present study, RAW264.7 macrophages were used as an in vitro cell model to investigate the immune modulation effect of HEP and hHEP. As shown in Fig. 2, the result showed that HEP and hHEP could significantly promote the proliferation rate of macrophages.

Macrophages have phagocytosis, which could phagocytose foreign substances such as pathogenic bacteria, acquire immunity through the process of presentation of antigen information on the surface of macrophages, and induct the specific antibody production and so on [28]. By staining dextran with FITC, the phagocytic properties of HEP and hHEP can be rapidly analyzed by flow cytometry (Fig. 4). In this study, HEP and hHEP are able to activate macrophages, which enhanced the vitality and increased the phagocytic function. Compared to HEP groups, hHEP (100 and 200 μg·mL^{-1})

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increase significantly higher phagocytosis of the RAW264.7 cells ($P < 0.05$). Recently, there have been several reports demonstrating that platycodon grandiflorus polysaccharides and yam mucopolysaccharides significantly enhanced the phagocytosis of RAW264.7 cells in a dose-dependent manner [29,30].

NO as an important effect of macrophage molecules, which could activate macrophages and kill pathogenic microorganisms and tumor cells [31-33]. At the same time, NO could enhance the lysis and phagocytosis of macrophages, playing an important role in regulating the body's immune [34]. Thus, the ability to release NO by macrophage can be detected and reflected that the effect of polysaccharides and their derivatives on immune function. In this study, HEP and hHEP can increase NO production and significantly higher than the control group ($P < 0.05$) (Fig. 3). Moreover, the hHEP at 200 $\mu g \cdot mL^{-1}$ was significantly increased higher than those in other groups ($P < 0.05$) and nearly identical with that of LPS group. This result is consistent with the similar immunomodulatory behavior of previously reported another polysaccharides [35, 36].

In the stimulation of certain substances, macrophages can activate transcription gene rapidly synthesis of iNOS mRNA. The iNOS mRNA translates expression of iNOS protein that catalyzes the production of a high concentration of NO [37,38]. Therefore, NO more than normal physiological concentration in the body is thought to be the result of the activation of the iNOS [39]. As shown in Fig. 6, the result showed that HEP and hHEP groups could significantly promote the iNOS expression than control groups ($P < 0.05$). Compared to HEP groups, hHEP (100 and 200 $\mu g \cdot mL^{-1}$)

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groups were significantly increased higher iNOS expression ($P < 0.05$). The results of the iNOS expression experiment were consistent with the previous NO production. These findings suggest that hHEP enhanced iNOS expression and immune activities.

Cytokines are small molecular proteins secreted by activated monocyte-macrophages and lymphocytes [40]. It can mediate and regulate immune response. IL-6 and TNF-α are an important cytokine produced by activated macrophages. As an important regulator of host defense responses, IL-6 contributes to the pathogenesis of various inflammatory and immunologic diseases [41, 42]. TNF-α could activate macrophages to enhance various functional responses and induce the expression of anti-tumor and immune regulatory mediators [43]. As shown in Fig. 5, Compared to HEP groups, hHEP at 200 $\mu g \cdot mL^{-1}$ group were significantly increased higher IL-6 and TNF-α production ($P < 0.05$). Similar results concerning about Actinidia eriantha polysaccharides and Ganoderma lucidum polysaccharides could effectively activate macrophages to release IL-6 and TNF-α [27, 44].

5. Conclusion

In this study, a homogeneous H. erinaceus polysaccharides were modified to obtain its hydroxyethylation derivatives, hHEP. It is the first report on the evaluation of the composition and immunomodulatory effects of hydroxyethylated polysaccharides from H. erinaceus. We have shown that hHEP promoted the phagocytic function. Moreover, hHEP showed stronger immunomodulatory activities than HEP in term of stimulating the production of NO and cytokines. Finally, hHEP enhanced the iNOS expression. The results suggested that hydroxyethylation

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modification can enhance the immune activity of HEP and also have the potential to become a valuable candidate for immunomodulatory agents.

Conflict of interests

Authors declare that there is no conflict of interests.

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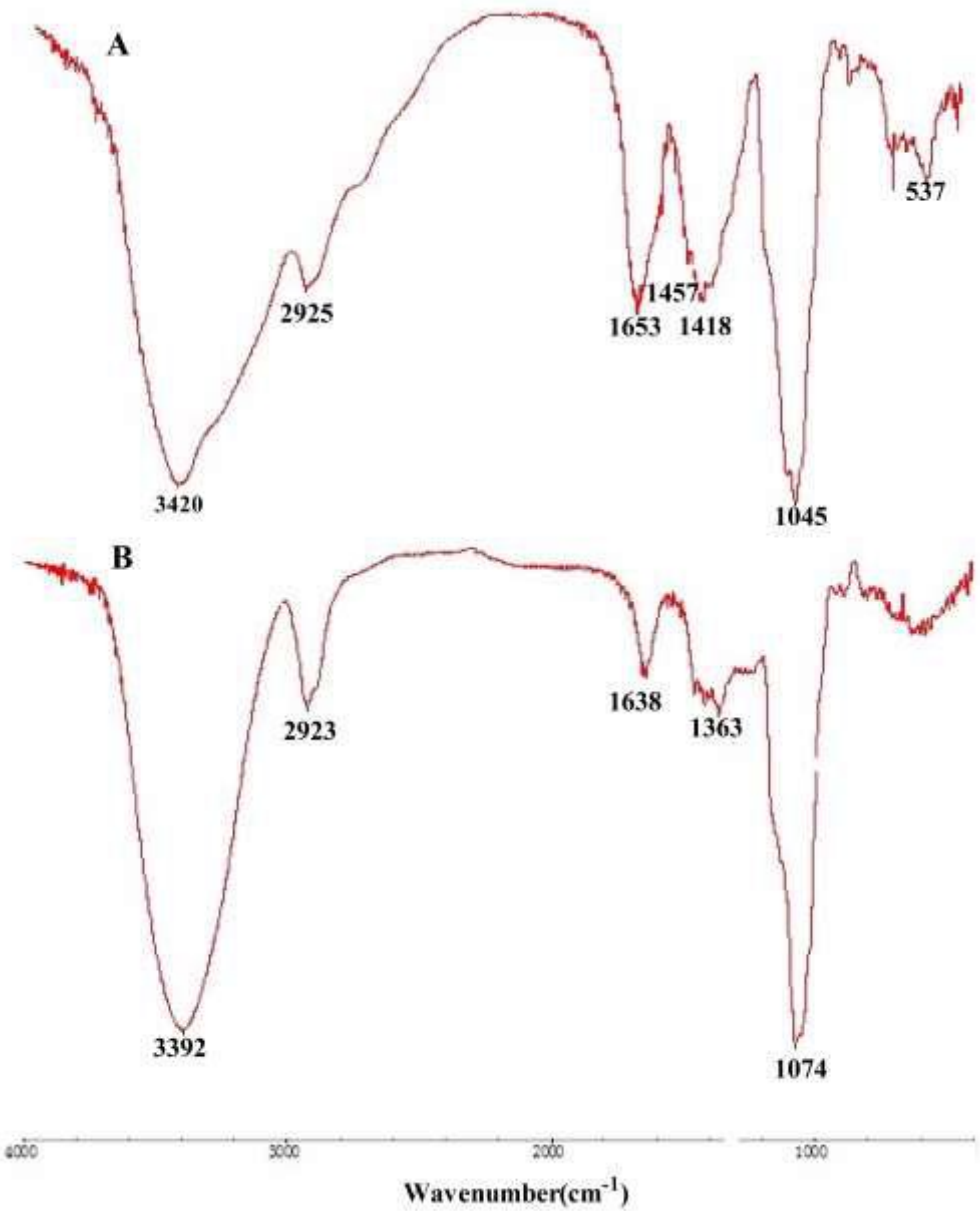

Fig. 1. (A) Infrared spectrum of HEP; (B) Infrared spectrum of hHEP.

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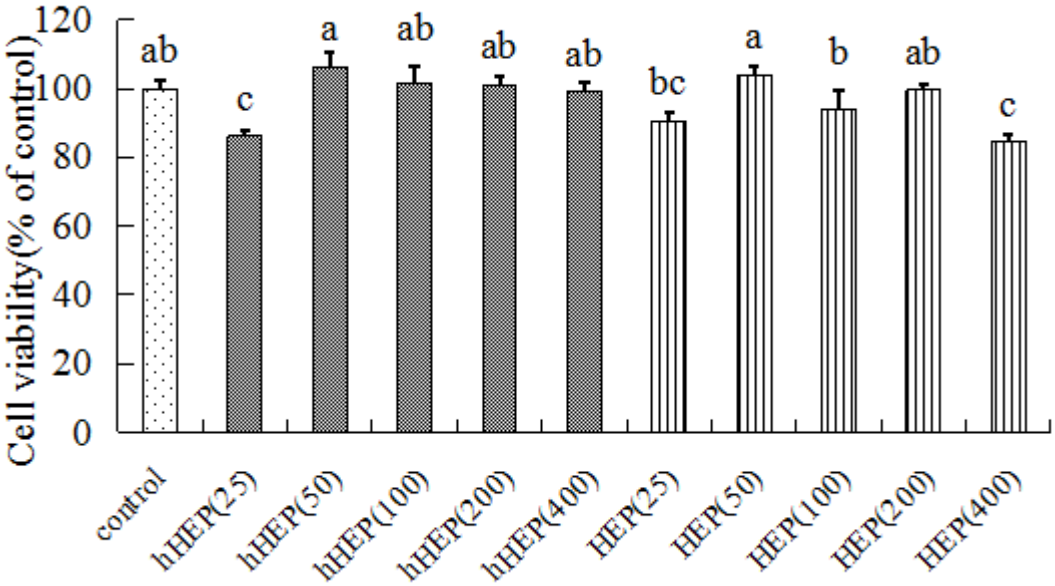

Fig. 2. Effects of HEP and hHEP on the viability of RAW264.7 cells. Column data marked without the same superscripts (a-c) differ significantly ($P < 0.05$).

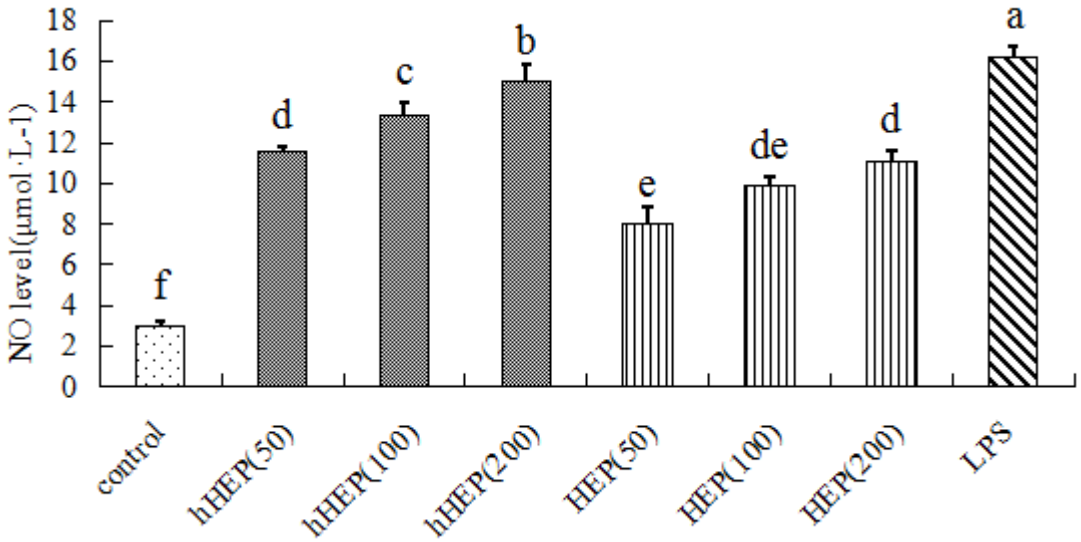

Fig. 3. Effect of HEP and hHEP on NO release from RAW264.7 cells. Column data marked without the same superscripts (a-f) differ significantly ($P < 0.05$).

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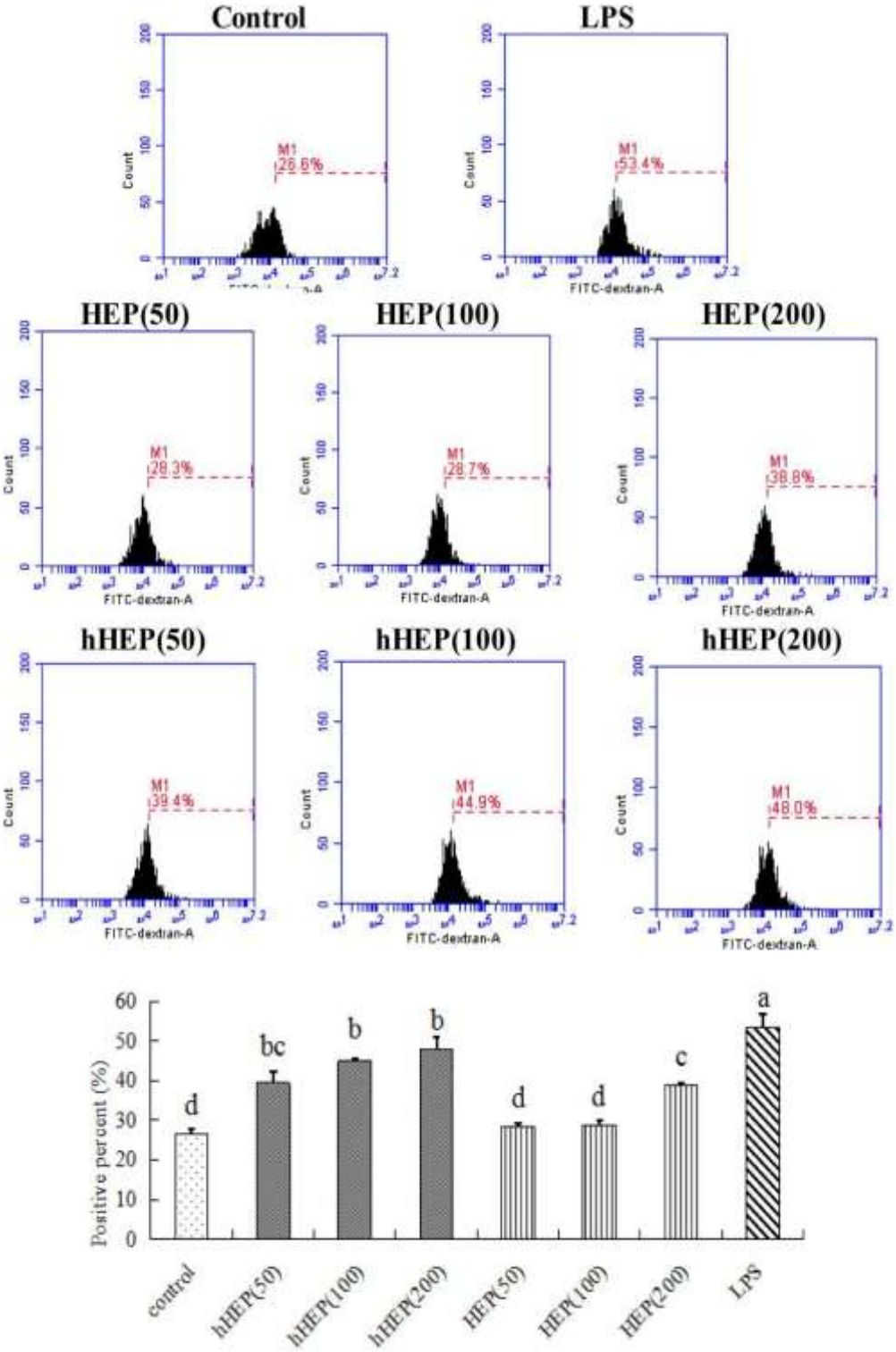

Fig. 4. Effect of HEP and hHEP on phagocytic uptake from RAW264.7 cells. The value are presented as mean ± SD (n = 4). Column data marked without the same superscripts (a-d) differ significantly ($P < 0.05$).

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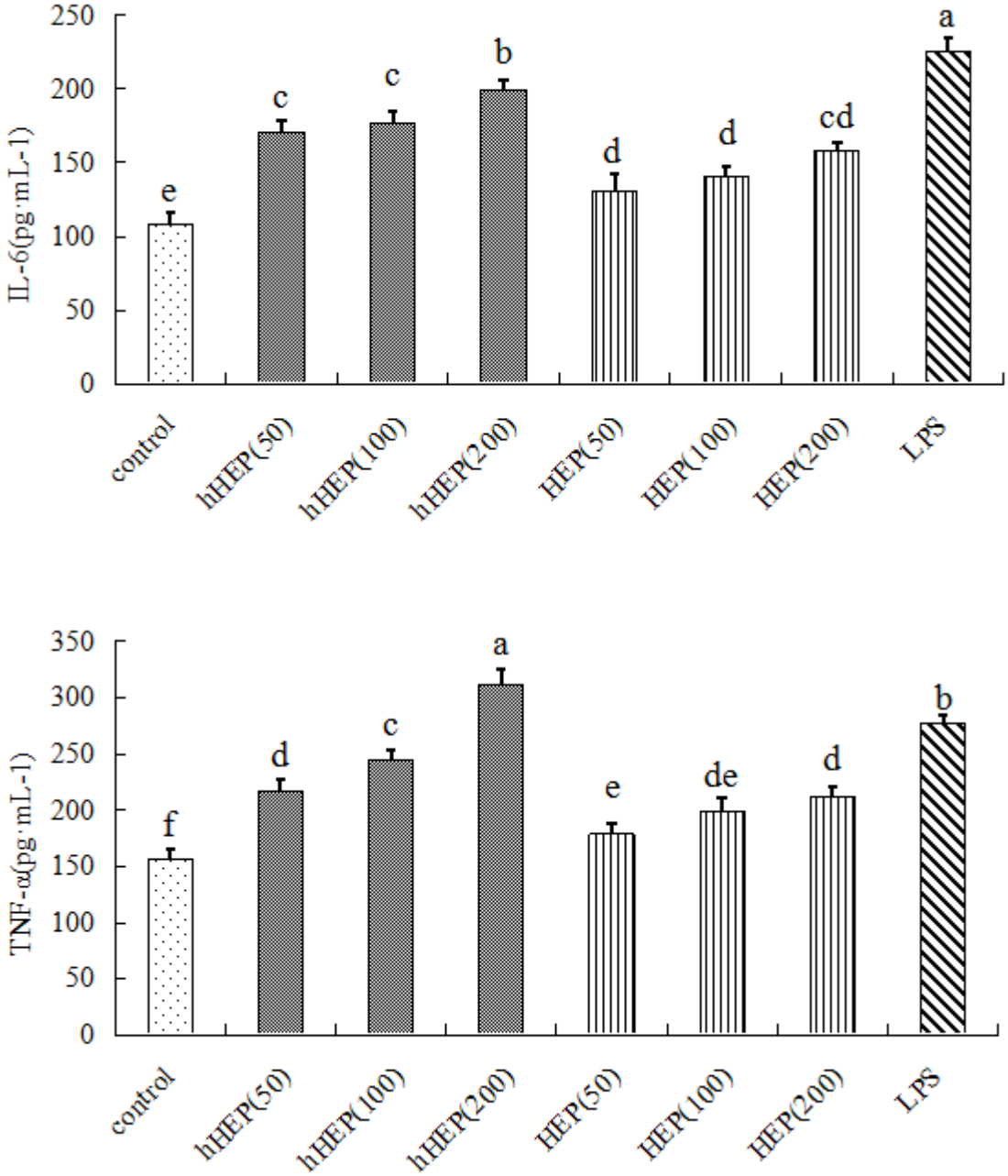

Fig. 5. Effect of HEP and hHEP on IL-6 and TNF-α from RAW264.7 cells. Column data marked without the same superscripts (a-f) differ significantly ($P < 0.05$).

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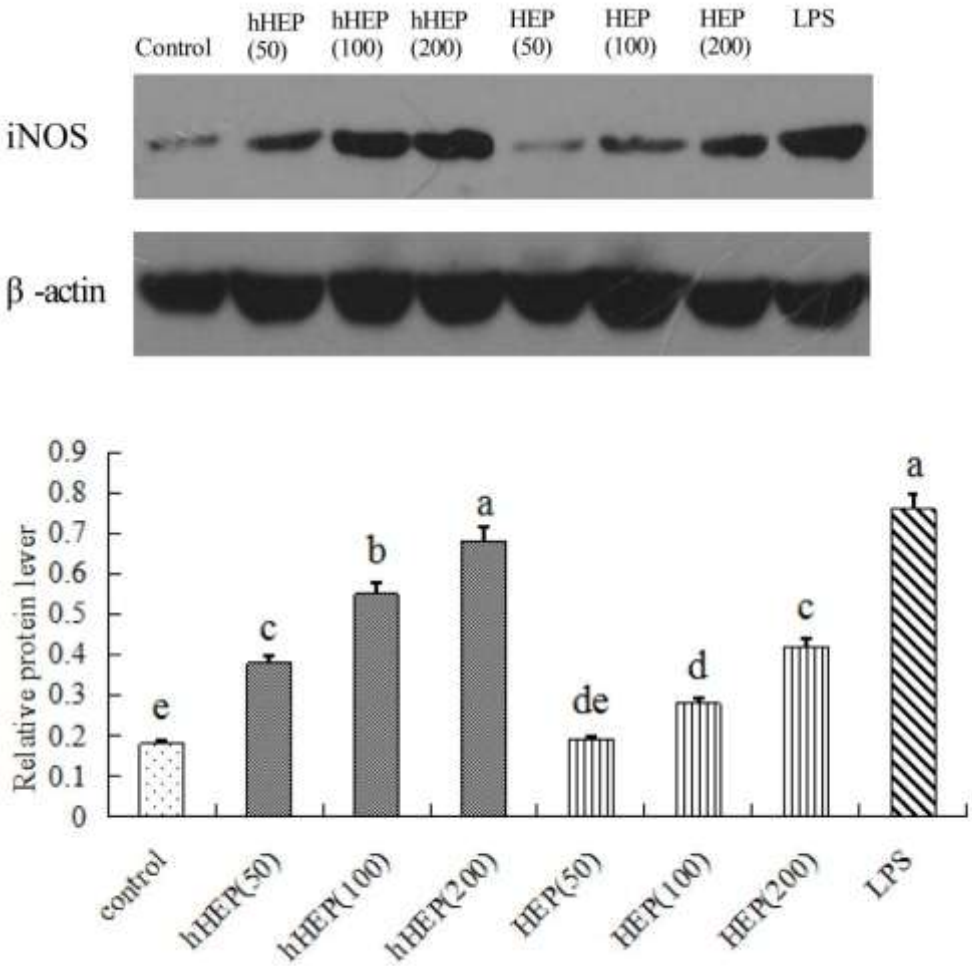

Fig. 6. Effect of HEP and hHEP on the iNOS protein expression level of RAW264.7 cells. Column data marked without the same superscripts (a-e) differ significantly ($P < 0.05$).

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Table 1

Chemical composition of HEP and hHEP

Sample	Mw(Da)	Carbohydrate(%)	Monosaccharide compositions
HEP	16.15 kDa	81.3 %	glucose, galactose, mannose and arabinose with a molar ratio of 50.80: 42.30: 4.58: 2.29
hHEP	15 kDa	75.4%	glucose, fructose, galactose, mannose and arabinose with a molar ratio of 30.23: 25.30: 2.45: 1.04: 0.18.